Younghoo Henry Cho

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Summary

Engineer who builds the infrastructure that makes AI accountable, so organizations can trust what AI tells them and verify why. Seven years across aquaculture production AI, agricultural data infrastructure, and cross-organizational distributed systems, taking ownership from problem definition through production deployment. Two filed patents and peer-reviewed publications.

Technical Skills

Backend & Distributed Systems: Hyperledger Fabric, Go, Node.js, Express, REST APIs, gRPC, microservices
Data & Pipelines: Apache Airflow, Apache Spark, Apache Kafka, data lineage, ETL, PostgreSQL, MongoDB, CouchDB, Firebase
Cloud & DevOps: AWS, Azure, Docker, Kubernetes, CI/CD (GitHub Actions), Git, Linux
Applied AI & Vision: LLM integration, agentic systems, prompt engineering, evaluation frameworks, PyTorch, YOLO, SAM, OpenCV
Languages: Python, Go, JavaScript, Java, C, C++, R

Experience

Graduate Research Assistant, University of Florida — **Aug 2022 – Aug 2026**

- Architected and shipped DEMETER end-to-end — a cross-organizational agricultural data verification platform — taking full ownership from system design through production deployment across four independently operated organizations.
- Built a distributed-ledger backend on Hyperledger Fabric (Go) with a Node.js/Express REST API service layer, enabling real-time data lineage tracing across 8,000+ linked records with sub-5s latency.
- Designed scalable event-driven ingestion pipelines using Airflow orchestration, Kafka event streams, and Spark batch aggregations over file-level metadata.
- Integrated a local LLM as a natural language query interface with a 15-intent classification system, deterministic routing, and a confirmation gate to prevent unverified writes — validated across a 74-case evaluation suite with zero false writes.
- Delivered production-grade documentation and demo code independently adopted by external collaborators across four organizations without direct handoff support.

Researcher, Ministry of Agriculture, Food and Rural Affairs (Korea) — **2024 – 2025**

- Conducted end-to-end technical evaluation of agricultural mechanization systems across South Korea and the United States, assessing field data requirements and technology adoption gaps to inform cross-country policy decisions.

AI/ML Engineer, Billion21 — **2019 – 2022**

- Built and deployed production computer vision pipelines (YOLO, SAM, OpenCV) for real-time marine monitoring and automated feeding, reducing manual inspection effort in live commercial aquaculture operations.
- Designed and shipped detection and segmentation models from prototype to production; core components contributed to two filed patents covering blockchain-tracked seaweed farming and spore selection.

Undergraduate Research Assistant, Texas Tech University — **2021 – 2022**

- Developed computer vision models for biological specimen detection and classification (plant species, animal behavior, acoustic signals), and built data analysis and visualization pipelines adopted by subsequent lab projects.

AI Technology Consultant, Global BioAg Linkages — **2020 – 2021**

- Evaluated AI technology fit for agricultural production systems end-to-end — from technical feasibility assessment and data requirement scoping to market viability analysis — producing deployment recommendations adopted in downstream pilot projects.

Education

University of Florida — Ph.D., Agricultural and Biological Engineering, GPA: 3.95/4.0 — **Aug 2022 – Expected Aug 2026**
Texas Tech University — B.S., Computer Science, Minor in Mathematics, GPA: 3.8/4.0 — **Aug 2015 – Jul 2022**

Selected Publications

- Cho, Y., Yu, Z., & Ampatzidis, Y. ***DEMETER: A Blockchain-Based Platform for Tracking the History of Shared Agricultural Data Files***. Under review.
- Cho, Y., Yu, Z., & Ampatzidis, Y. ***Facilitating a Future Agricultural Data Ecosystem: A Cross-Sectoral Review of Blockchain Application in Data Sharing***. Under review.
- Cho, Y., Yu, Z., & Ampatzidis, Y. (2025). ***Bridging Carbon Markets and Agriculture: A Comparative Study of Stakeholder Engagement in South Korea and United States***. *Discover Agriculture*, 3, 126.

Patents

- ***Blockchain-Based Seaweed Farming Data Management and Smart Supply Chain Tracking System***. KR10-2024-0011399, 2024. **Status: Pending.**
- ***Methods for Selecting Spores from Laver Farm Devices***. KR102034354B1, 2019. **Status: Granted.**